Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 233 (2024) 832–840

Procedia Computer Science

www.elsevier.com/locate/procedia

5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

Automated Bone Fracture Detection in X-ray Imaging to Improve Orthopaedic Diagnostics in Healthcare

Soumi Ghosh[a], Jahiruddin Mallick[a], Monalisa Chakraborty [a], Subir Gupta [b]

[a]*Dr. B. C. Roy Engineering College, Jemua Road, Fuljhore, Durgapur – 713206, India*
[b]*Dept. of C.S.E. Swami Vivekananda University, Kolkata, West Bengal, India*

Abstract

The healthcare system in India faces an acute shortage of medical staff, significantly impacting the nation's ability to provide sufficient healthcare services. X-ray reports are paramount to orthopaedic surgeons for making critical patient care decisions. Implementing automated systems for detecting bone fractures offers a potentially efficient solution to supplement the limited workforce. However, these systems encounter substantial hurdles, such as inconsistent image quality and the risk of incorrect diagnosis due to the limitations of current technology. The research provides a specialised tool for identifying bone fractures with high precision (97% accuracy) and an excellent overall rating (F1 score of 0.98), showing it can reliably detect fractures. The assessment of this model is augmented by visual tools like heatmaps and performance graphs, providing an in-depth analysis of its operational effectiveness. Furthermore, creating a user-friendly interface is intended to streamline the adoption of this tool in clinical settings. Notwithstanding its demonstrated precision, the model must navigate data-related challenges, precisely data complexity and the representation of different categories, to prevent skewed outcomes. Clarifying the model's decision-making process and adaptability to various clinical situations are also critical. The study emphasises the need for thorough testing to ensure the model's dependability in diagnosing fractures before being deployed effectively in the medical field.

© 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

Keywords: Bone Fracture ; X-ray Imae; Healthcare ; Machine Learning ;

1. Introduction

India's healthcare infrastructure is facing a severe crisis due to a shortage of trained healthcare workers, affecting all levels, from doctors and specialists to nurses and lab technicians[1]. This issue is particularly acute in rural areas and less affluent urban neighbourhoods, exacerbating the divide in access to healthcare services[2]. This shortfall is impeding the effectiveness of the healthcare system, delaying the provision of essential medical care to India's vast

1877-0509 © 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application
10.1016/j.procs.2024.03.272

and growing population. Addressing this critical situation requires innovative solutions, including expanding the medical workforce and integrating modern technology to fill the current void. In diagnostic medicine, and orthopaedics in particular, X-ray imaging is a fundamental diagnostic tool. It provides detailed views of the skeletal system, revealing conditions such as bone fractures, joint dislocations, and signs of arthritis. Accurate interpretation of X-ray images is essential for setting correct treatment plans, tracking recovery, and deciding on further medical or surgical steps[3][4].

Nonetheless, the lack of enough qualified radiologists, especially in certain regions, limits the full use of this essential diagnostic method. In response to these challenges, the study presents a new approach based on advanced computing techniques to automate the identification of bone fractures from X-ray images. This method employs sophisticated software trained on an extensive collection of X-ray data to mimic—and potentially exceed—the diagnostic capabilities of human experts. Such a technique promises to speed up the diagnosis process and improve the consistency and accuracy of fracture identification, which is especially beneficial in settings where radiologists are in short supply[5][6]. This study pinpoints a dual issue, the shortage of medical staff and the underuse of diagnostic tools, as a significant gap in healthcare service delivery. The proposed approach aims to tackle this gap, providing a practical, precise, and scalable option compared to current diagnostic methods, thereby improving patient care and the efficient use of medical resources. The remaining text of the study is as follows: In Section 2, study the literature review. Section 3 will describe the system The study recommends, while Section 4 will describe the findings in the last segment.

2. Literature Review

The development of automated methods for identifying bone fractures through X-ray imaging has emerged as a significant advancement in healthcare, especially for diagnosing orthopaedic conditions[7][8]. The academic discourse on this topic is rich and varied, blending progress in image processing techniques, the rise of advanced computational methods, and the urgent requirement for such innovations due to a shortage of healthcare professionals. Initial studies focused on conventional image processing methods for fracture detection, employing techniques like edge detection, but these were often inadequate due to the diverse nature of fractures. Advances in computational methods, especially Convolutional Neural Networks (CNNs), have led to more sophisticated image analyses. Research has demonstrated that CNNs, when trained on extensive and varied datasets, have the potential to surpass older methods in detecting fractures, including those that are not immediately obvious to the human eye. The concept of transfer learning has been pivotal in mitigating the challenges associated with the scarcity of medical imaging data[9].

By adapting models trained on generic images to medical tasks, researchers have found that they can achieve significant improvements without needing large, specialised medical datasets. The significance of high-quality training data is crucial for the success of these models[10]. The push for creating comprehensive, annotated datasets that reflect real-world complexities has been a significant focus, with initiatives like the DICOM database making substantial contributions to this effort [11]. The interpretability of these computational models is a critical factor in their clinical adoption. Research into visualisation techniques, such as Class Activation Mapping, has been conducted to provide insights into the decision-making processes of these models, enhancing the trust of medical professionals in the technology. Performance evaluation of automated systems has been a topic of extensive discussion, with studies advocating for consistent accuracy across different patient demographics and imaging technologies. Rigorous validation efforts have been undertaken to ensure the reliability of these models. The challenge of integrating these automated systems into the existing healthcare IT infrastructure has been explored, emphasising the need for seamless integration to encourage widespread adoption. Deploying such technologies has also addressed data security and patient privacy concerns. The potential for automated tools to displace radiologists has been discussed, with some experts arguing that these tools should complement rather than replace human expertise. Ethical considerations, particularly the legal implications of relying on machine-based diagnostics, have been debated, with proposals for ethical frameworks to guide the deployment of these systems[12]. Finally, the potential for computational models to alleviate the shortage of healthcare workers has been underscored, with calls for the accelerated development of such systems in the face of global healthcare workforce shortages[13][14]. In conclusion, the body of literature points towards a shift from basic image processing to sophisticated computational

models for fracture detection. While promising, the full-scale implementation of these systems must address technical, ethical, and societal issues to ensure they are reliable, understandable, and ethically sound within the dynamic field of medical diagnostics[15].

3. Methodology

The methodology of this research article unfolds with the collection of X-ray images, which constitute the foundational data for the study. This essential first step involves compiling a dataset that showcases a wide array of case types, which is crucial for practical CNN training. Researchers source these images from trustworthy and ethical repositories that include a spectrum of pathological conditions. After gathering the images, preprocessing becomes the priority to render the images suitable for analysis. This critical phase entails applying filters to augment the images' features and resizing them to create a standard input size for the CNNs. For instance, edge detection filters may be used to accentuate anatomical structures. The resizing process carefully preserves the aspect ratio, thus avoiding feature distortion. Labelling the images is the next critical task, significantly influencing the model's training effectiveness. Experts categorise each image according to texture, shape, and frequency characteristics. Texture labelling classifies images by surface feature appearance. Shape labelling deals with the images' geometric features, while frequency labelling addresses the spatial frequency of pixel intensities.

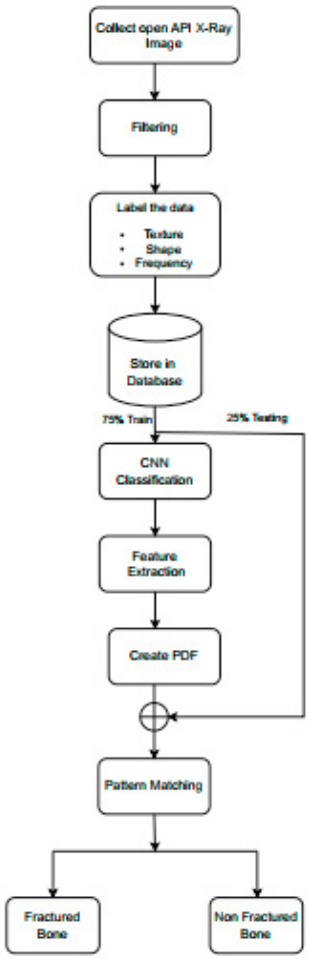


Figure 1: Methodology Diagram

Expert radiologists perform this labelling to guarantee label precision, vital for CNN's supervised learning. The proposed model then split the dataset into training and testing subsets at a 3:1 ratio, dedicating three-quarters to CNN model training and one-quarter to performance testing.

Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 128, 128, 3)	0
conv2d (Conv2D)	(None, 128, 128, 16)	448
max_pooling2d (MaxPooling2D)	(None, 64, 64, 16)	0
conv2d_1 (Conv2D)	(None, 64, 64, 32)	4640
max_pooling2d_1 (MaxPooling 2D)	(None, 32, 32, 32)	0
conv2d_2 (Conv2D)	(None, 32, 32, 64)	18496
max_pooling2d_2 (MaxPooling 2D)	(None, 16, 16, 64)	0
dropout (Dropout)	(None, 16, 16, 64)	0
flatten (Flatten)	(None, 16384)	0
dense (Dense)	(None, 128)	2097280
dense_1 (Dense)	(None, 2)	258

Total params: 2,121,122
Trainable params: 2,121,122
Non-trainable params: 0

Figure 2: Dense Layer

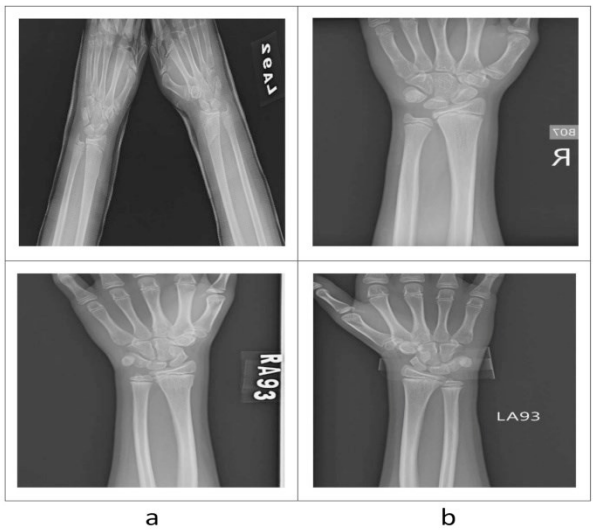


Figure 3: Sample images

This strategic division ensures the model can generalise to new data and maintains a balanced representation of different cases in both subsets. Training the dataset involves inputting it into a CNN model, which emulates the neural connectivity of the human brain, allowing it to identify patterns with notable accuracy. In this study, the model undergoes 30 training epochs, with one epoch representing one complete cycle of the training dataset through the network. During these epochs, the team adjusts the model's parameters to reduce prediction errors. After training completion, the team develops a probability density function (PDF) for the prediction phase. This PDF estimates the probability of new X-ray images belonging to the learned categories. Figure 3 shows the sample images. It is essential to the model's predictive capacity, providing probabilistic outcomes for meaningful interpretation. Documenting each step with precision is mandatory in the methodology section. The researchers elucidate every phase in detail, from data collection through model training to prediction, describing the software, algorithms, training parameters, and statistical methods employed. Such thorough documentation ensures the study is replicable and provides a foundation for further research—an essential aspect of scientific progress. Figure 1, the study can see the methodology diagram of this process, and Figure 2 displays the Dense Layer architecture of a CNN model.

4. Result

In the study, the detailed results of a deep learning model for bone fracture detection, highlighted by an F1 score of 0.98 and an accuracy of 0.97, demonstrate a high level of performance. Figure 5 illustrates a graph of a model's accuracy. These metrics suggest that the model is exceptionally proficient in identifying true positives while minimising false negatives and false positives, a critical achievement in medical diagnostics where the consequences of incorrect predictions can be severe. Despite being numerically close, the true positive rate of 15 and the false negative rate of 17 are particularly salient in the context of a binary classification task, such as bone fracture detection.

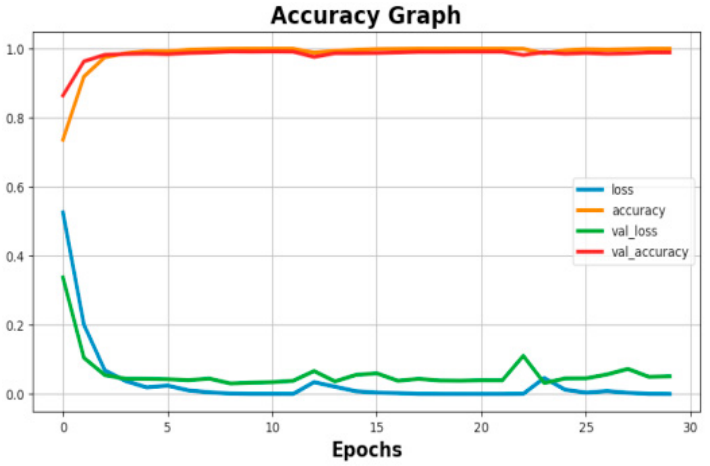


Figure 4: Accuracy Graph

This proximity raises questions about the model's sensitivity, a measure of its ability to identify true positives. A high sensitivity is paramount in medical applications to ensure that conditions are not overlooked. The high F1 score indicates that the model maintains a balanced precision-recall relationship, which, in the context of medical imaging, translates to correctly identifying fractures when they are present and correctly ruling them out when they are absent. Figure 7 depicts picking parameters using a heat map. Providing heatmaps, accuracy, and log-loss graphs enriches the study by offering visual and quantitative insights into the model's performance across different thresholds and epochs. Heatmaps are particularly useful in medical imaging to pinpoint the areas the model identifies as indicative of fractures, offering interpretability to the model's decision-making process. Accuracy graphs over epochs can illustrate the model's learning trajectory, providing evidence of stable learning or

highlighting potential overfitting. Figure 4 shows the accuracy graph. Log-loss graphs, or cross-entropy loss graphs, indicate the predictions' confidence, with lower values suggesting higher confidence levels. Figure 6, which may be seen further down in this section for showing the loss graph. The test validation is shown in Figure 8.

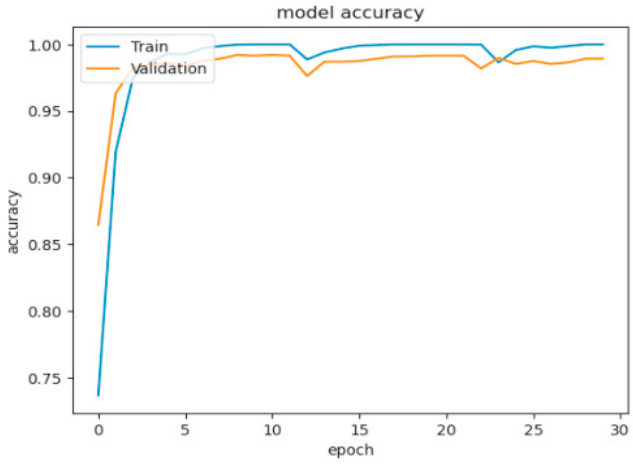


Figure 5: Accuracy Model

Implementing a Graphical User Interface (GUI) for the application reflects the study's commitment to translational research, bridging the gap between high-level technical innovation and practical clinical utility. A user-friendly GUI is essential for adoption in clinical settings, allowing healthcare professionals to interact with the technology effectively. However, the model's effectiveness must be interpreted in the context of the dataset's composition. A binary dataset for bone fracture detection typically includes images labelled 'fractured' or 'not fractured.' The class distribution within this dataset can significantly impact model performance metrics. If there is a significant imbalance, with one class being overrepresented, it can lead to a biased model that may not perform equally well on a more balanced real-world dataset. Graphical User Interface are shown in Figure 9.

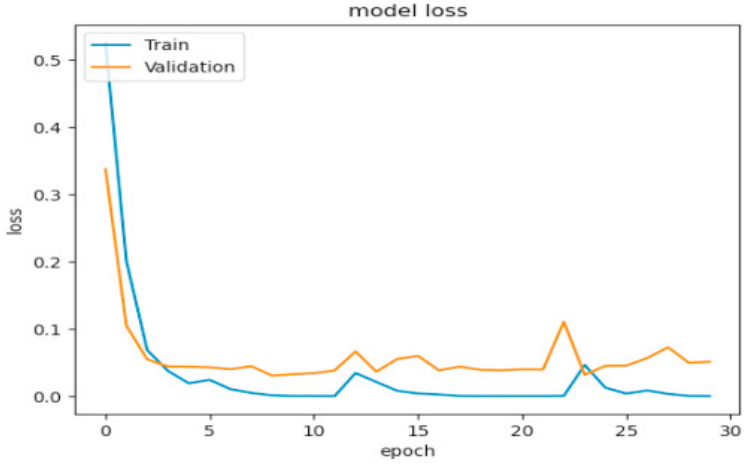


Figure 6: Loss Graph

Further, the article should discuss the dimensionality of the feature space and how the model handles class imbalances, if present. Dimensionality reduction techniques or class-weighting mechanisms could improve model performance and ensure it does not become biased towards the majority class. For the research to have a significant

impact, it must also address the potential biases present in the dataset and how they were mitigated. The interpretability of the model's predictions is crucial, especially in healthcare applications where decisions have significant consequences. The model's scalability to different datasets and real-world scenarios is another critical factor that would determine the model's utility beyond the research environment.

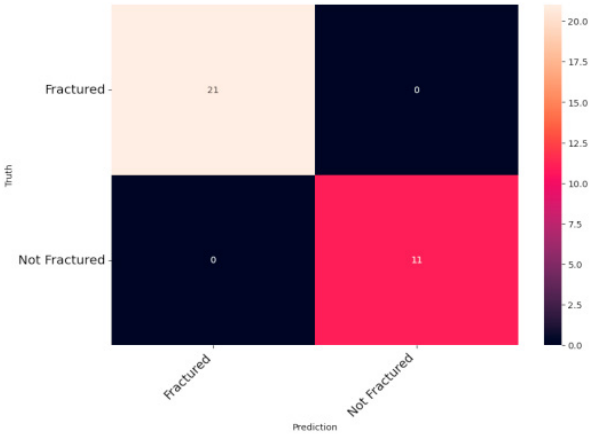


Figure 7: Heatmap Graph



Figure 8: Validation of the proposed methodology



Figure 9: GUI of the proposed methodology

In summary, the results suggest a highly effective model for bone fracture detection, as evidenced by the high F1 score and accuracy. Including a GUI and various performance graphs demonstrates a comprehensive approach to model development and evaluation. Nonetheless, for these results to be deemed impactful in the broader scientific community, the research must critically evaluate the dataset's composition, address potential biases, ensure model interpretability, and demonstrate scalability. Only by thoroughly analysing these factors can the research claim to offer a model that achieves high performance and is ready for real-world clinical application.

5. Conclusion

In summary, this research details a proficient deep-learning model for detecting bone fractures, achieving a high F1 score of 0.98 and an accuracy of 0.97. Such a model is critical for the medical field, where precise diagnosis is paramount. The reported true positive and false negative rates suggest a robust balance in the model's ability to correctly identify fractures, a vital aspect of a binary classification task in medical imaging. Visual tools like heatmaps and performance graphs offer in-depth views of the model's functionality, aiding in assessing its predictive confidence and learning consistency. Heatmaps are particularly beneficial in highlighting the regions identified by the model as potential fractures, increasing the interpretability of their outputs. The graphical representation of the model's accuracy and log-loss over multiple epochs also helps identify any overs fitting signs, enhancing trust in its reliability. The study's development of a Graphical User Interface (GUI) showcases a commitment to making technical advancements accessible and practical for clinical use. The GUI's design is integral to its adoption by healthcare professionals, facilitating seamless interaction with the technology.

Nonetheless, considering the dataset's structure, the model's performance must be considered. The binary classification of 'fractured' or 'not fractured' and the distribution of these classes can significantly influence the model's effectiveness. Techniques to manage dimensionality and class imbalance are vital to ensure the model's unbiased functioning and to maintain its high performance across varied datasets. Furthermore, the model's interpretability and scalability are essential for real-world applications. It must perform reliably across different datasets and practical clinical scenarios to be truly impactful. Overall, the research indicates a sophisticated and potentially transformative tool for fracture detection, pending a critical evaluation of the dataset, bias mitigation, and verification of the model's interpretability and generalizability. This holistic analysis is crucial to affirming the model's readiness for clinical deployment and cementing its significance in the scientific community.

References

[1] A. T. Almaktoom, “Health care overbooking cost minimization model,” *Heliyon*, vol. 9, no. 8, p. e18753, 2023, doi: 10.1016/j.heliyon.2023.e18753.

[2] T. Åsgård and H. Breistrand, “ScienceDirect ScienceDirect ScienceDirect Co-determination and Information in Information Experiences from the Information construction of a hospital building Health and Social 2022 in Co-determination and participation Norway . in project management. an,” *Procedia Comput. Sci.*, vol. 219, no. 2022, pp. 1744–1751, 2023, doi: 10.1016/j.procs.2023.01.469.

[3] D. S. S. K. Vipin Y. Borole, Sunil S. Nimbhore, “Image Processing Techniques for Brain Tumor Detection: A Review,” *Int. J. Emerg. Trends Technol. Comput. Sci. Web*, vol. 4, no. 5, pp. 28–32, 2015, [Online]. Available: http://www.ijettcs.org/Volume4Issue5(2)/IJETTCS-2015-10-01-7.pdf

[4] B. Sekeroglu and I. Ozsahin, “Detection of COVID-19 from Chest X-Ray Images Using Convolutional Neural Networks,” *SLAS Technol.*, vol. 25, no. 6, pp. 553–565, 2020, doi: 10.1177/2472630320958376.

[5] A. E. Rodríguez-Soto *et al.*, “Texture Analysis, Bone Mineral Density, and Cortical Thickness of the Proximal Femur,” *J. Comput. Assist. Tomogr.*, vol. 34, no. 6, pp. 949–957, Nov. 2010, doi: 10.1097/RCT.0b013e3181ec05e4.

[6] A. Cheng, Q. Guan, Y. Su, P. Zhou, and Y. Zeng, “Integration of Machine Learning and Blockchain Technology in the Healthcare Field: A Literature Review and Implications for Cancer Care,” *Asia-Pacific J. Oncol. Nurs.*, vol. 8, no. 6, pp. 720–724, 2021, doi: 10.4103/apjon.apjon-2140.

[7] J. Sun, X. Li, C. Tang, S. H. Wang, and Y. D. Zhang, “MFBCNNC: Momentum factor biogeography convolutional neural network for COVID-19 detection via chest X-ray images[Formula presented],” *Knowledge-Based Syst.*, vol. 232, p. 107494, 2021, doi: 10.1016/j.knosys.2021.107494.

[8] S. Asif, Y. Wenhui, H. Jin, and S. Jinhai, “Classification of COVID-19 from Chest X-ray images using Deep Convolutional Neural Network,” *2020 IEEE 6th Int. Conf. Comput. Commun. ICCC 2020*, pp. 426–433, 2020, doi: 10.1109/ICCC51575.2020.9344870.

[9] L. C. Ngugi, M. Abelwahab, and M. Abo-Zahhad, “Recent advances in image processing techniques for automated leaf pest and

disease recognition – A review," *Inf. Process. Agric.*, vol. 8, no. 1, pp. 27–51, 2021, doi: 10.1016/j.inpa.2020.04.004.

[10] R. P. *et al.*, "Deep learning for chest radiograph diagnosis: A retrospective comparison of the CheXNeXt algorithm to practising radiologists.," *PLoS Med.*, vol. 15, no. 11, p. e1002686, 2018, [Online]. Available: https://pubmed.ncbi.nlm.nih.gov/30457988/%0Ahttp://medicine.plosjournals.org/perlserv/?request=index-html&issn=1549-1676

[11] D. Demner-Fushman *et al.*, "Preparing a collection of radiology examinations for distribution and retrieval," *J. Am. Med. Informatics Assoc.*, vol. 23, no. 2, pp. 304–310, 2016, doi: 10.1093/jamia/ocv080.

[12] K. Stokes *et al.*, "A machine learning model for supporting symptom-based referral and diagnosis of bronchitis and pneumonia in limited resource settings," *Biocybern. Biomed. Eng.*, vol. 41, no. 4, pp. 1288–1302, 2021, doi: 10.1016/j.bbe.2021.09.002.

[13] M. Adil and M. K. Khan, "Emerging IoT Applications in Sustainable Smart Cities for COVID-19: Network Security and Data Preservation Challenges with Future Directions," *Sustain. Cities Soc.*, vol. 75, no. June, p. 103311, 2021, doi: 10.1016/j.scs.2021.103311.

[14] C. Ma, "Smart city and cyber-security; technologies used, leading challenges and future recommendations," *Energy Reports*, vol. 7, pp. 7999–8012, 2021, doi: 10.1016/j.egyr.2021.08.124.

[15] M. Naveenkumar and V. Ayyasamy, "OpenCV for Computer Vision Applications," *Proc. Natl. Conf. Big Data Cloud Comput.*, no. March 2015, pp. 52–56, 2016, [Online]. Available: https://www.researchgate.net/publication/301590571_OpenCV_for_Computer_Vision_Applications